

This Assignment will not be considered for Course participation Marks after 16th Oct 2022

Instructions for Extended Match Questions

Use this slide to understand how to answer an EMQ

Instructions:

1. For each question, enter all the options which you find is the answer to that question as "**comma separated values**".
2. Enter the options in **ascending order** of option number.
3. An option can also be an answer to more than one question, or may not be an answer to any of the questions.
4. Please click [here](#) to view a sample template file for answering.
5. Please note that if you give an incorrect answer there will be a negative marking for the incorrect option.

List of Options:

- 1.Linearly independent set.
- 2.Linearly dependent set.
- 3.If we remove any one of the vectors from the set, then it will be linearly independent.
- 4.If we put the vector $(1, 1, 1)$ in the set, then it will be linearly dependent.
- 5.There exist one vector in the set, if we remove that vector from the set, then it will be linearly independent.
- 6.If we remove any two vectors from the set, then the set will be linearly independent.

List of Questions:

Q1.Consider the set $S_1 = \{(1, 0, 1), (1, 0, 0), (1, 1, 0), (0, 1, 0)\}$. Which of the above options is(are) true for S_1 ?

Q2.Consider the set $S_2 = \{(1, 1, 0), (1, 1, 0) + (0, 1, 1), (1, 1, 0) - (0, 1, 0)\}$ with usual coordinate-wise addition and scalar multiplication. Which of the above options is(are) true for S_2 ?

Q3.Consider the set of all points on X -axis, each one treated as a different vectors in $\mathbb{R}^3$, with usual addition and scalar multiplication. Which of the above options is(are) true for this set?